

CURIONEST

Buffers, Titration & Salt Hydrolysis

Practice Pack · Henderson–Hasselbalch · Equivalence Points · Salt Ions · Full Answer Key

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

What This Pack Covers

1. Calculating buffer pH with the Henderson–Hasselbalch equation
2. Predicting how buffers respond to added acid or base
3. Classifying salt solutions as acidic, basic, or neutral
4. Reading titration curves and locating equivalence points
5. Choosing appropriate indicators for titrations

Reading Passage — Buffers, Salts, and Titrations

A buffer contains a weak conjugate acid–base pair, such as CH_3COOH and CH_3COO^- . It resists pH change because added H^+ is consumed by the base partner and added OH^- is consumed by the acid partner. The Henderson–Hasselbalch equation, $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$, gives the buffer's pH; when $[\text{A}^-] = [\text{HA}]$, $\text{pH} = \text{pK}_a$.

Buffer capacity is the amount of acid or base a buffer can absorb before pH changes sharply. Capacity is largest when both partners are concentrated and roughly equal.

Salt ions can hydrolyze water: the conjugate base of a weak acid (e.g., CN^-) makes a solution basic; the conjugate acid of a weak base (e.g., NH_4^+) makes it acidic; ions from strong acids and bases (Cl^- , Na^+) are neutral.

In a titration, pH changes slowly in buffer regions and jumps sharply at the equivalence point. For a weak acid titrated with strong base, the equivalence-point pH is above 7, and the half-equivalence pH equals pK_a . An indicator must change color near the equivalence-point pH.

★ **Common mistake:** Using the Henderson–Hasselbalch equation for strong acids or strong bases — it only applies to weak conjugate buffers.

★ **Common mistake:** Assuming equivalence-point pH is always 7.

Quick Check — Multiple Choice (Questions 1–8)

1. A buffer is made from which pair?
 - (A) HCl and NaCl
 - (B) CH_3COOH and CH_3COONa
 - (C) HNO_3 and NaNO_3
 - (D) NaOH and KOH
2. The pH of a buffer equals pK_a when:
 - (A) $[\text{HA}] = 2[\text{A}^-]$
 - (B) $[\text{A}^-] = [\text{HA}]$
 - (C) $[\text{A}^-] = 2[\text{HA}]$
 - (D) the buffer is diluted
3. Which salt solution is BASIC?
 - (A) NH_4Cl
 - (B) KCN
 - (C) NaCl
 - (D) $\text{Al}(\text{NO}_3)_3$
4. Which salt solution is ACIDIC?
 - (A) KCl
 - (B) NaCH_3COO
 - (C) NH_4Br
 - (D) RbI
5. In the titration of a weak acid with a strong base, the equivalence-point pH is:
 - (A) Exactly 7.00
 - (B) Below 7
 - (C) Above 7
 - (D) Equal to pK_a

6. At the half-equivalence point of a weak-acid titration, $\text{pH} = \text{pK}_a$ because:
- (A) All acid is neutralized
 - (B) $[\text{HA}] = [\text{A}^-]$
 - (C) $[\text{H}_3\text{O}^+] = [\text{OH}^-]$
 - (D) The acid is strong
7. Adding 0.01 mol HCl to 1.0 L of a buffer with $[\text{CH}_3\text{COOH}] = [\text{CH}_3\text{COO}^-] = 0.50 \text{ M}$ changes pH by approximately:
- (A) 4 pH units
 - (B) 0.02 pH units
 - (C) 7 pH units
 - (D) 14 pH units
8. Which indicator suits a titration whose equivalence-point pH is about 8.7?
- (A) Methyl red (4.4–6.2)
 - (B) Phenolphthalein (8.2–10.0)
 - (C) Bromothymol blue (6.0–7.6)
 - (D) Methyl orange (3.1–4.4)

Calculate — Show Your Work (Questions 9–14)

9. Calculate the pH of a buffer with 0.40 M CH_3COOH and 0.40 M CH_3COONa ($K_a = 1.8 \times 10^{-5}$).
10. Calculate the pH of a buffer with 0.25 M NH_3 and 0.35 M NH_4Cl ($K_b = 1.8 \times 10^{-5}$).
11. A buffer contains 0.30 M HF and 0.30 M NaF ($K_a = 6.8 \times 10^{-4}$). What is the pH after 0.010 mol of HCl is added to 1.0 L of the buffer?
12. Calculate the pH at the half-equivalence point of the titration of 40 mL of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) with 0.100 M NaOH.
13. Calculate the pH at the equivalence point of the same titration (40 mL 0.100 M CH_3COOH + 40 mL 0.100 M NaOH).
14. Calculate the pH after 41 mL of 0.100 M NaOH is added to 40 mL of 0.100 M HCl.

Short Answer & Think (Questions 15–18)

15. Explain why a buffer made of equal concentrations of a weak acid and its conjugate base has its maximum buffer capacity at that point.

16. Determine whether aqueous solutions of FeCl_3 , K_2CO_3 , NH_4Br , and KClO_4 are acidic, basic, or neutral, and explain each.
17. Explain why the equivalence-point pH of a weak-base + strong-acid titration is below 7.
18. Why does an indicator change color over a range of pH rather than at a single pH value?

Key Terms

Term	Definition
buffer	Solution of a weak conjugate acid–base pair that resists pH change
Henderson–Hasselbalch equation	$\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$
buffer capacity	Amount of acid or base a buffer can neutralize before pH changes sharply
hydrolysis	Reaction of a salt ion with water producing H_3O^+ or OH^-
equivalence point	Titration added exactly neutralizes the analyte
half-equivalence point	Half the analyte neutralized; $\text{pH} = \text{pK}_a$
indicator	Weak acid whose color changes across a pH range

Answer Key

All answers with explanations are embedded in the questions above — use the practice sections for the complete key.

Quick Check — Multiple Choice (Questions 1–8)

1. CH_3COOH and CH_3COONa (a weak acid with its conjugate base)

Why: Buffers need a weak conjugate acid–base pair, not a strong acid with its salt.

2. $[\text{A}^-] = [\text{HA}]$ ($\log(1) = 0$)

Why: $\text{pH} = \text{pK}_a + \log([\text{A}^-]/[\text{HA}])$; when the ratio is 1, $\log = 0$.

3. KCN (CN^- is the conjugate base of weak acid HCN)

Why: CN^- hydrolyzes to produce OH^- . NH_4^+ and Al^{3+} make acidic solutions; NaCl is neutral.

4. NH_4Br (NH_4^+ is the conjugate acid of weak base NH_3)

Why: NH_4^+ donates a proton to water, producing H_3O^+ .

5. Above 7 (the conjugate base hydrolyzes to give OH^-)

Why: At equivalence, the solution contains the salt of a weak acid + strong base, which is basic.

6. $[\text{HA}] = [\text{A}^-]$ (half the acid converted to conjugate base)

Why: With equal concentrations of both partners, the Henderson–Hasselbalch log term is zero.

7. 0.02 pH units (pH goes from 4.74 to about 4.72)

Why: The buffer absorbs the H^+ : $[\text{A}^-] = 0.49$, $[\text{HA}] = 0.51 \rightarrow \text{pH} = 4.74 + \log(0.49/0.51) \approx 4.72$.

8. Phenolphthalein (8.2–10.0 brackets 8.7)

Why: The indicator's color-change range must include the equivalence-point pH.

Calculate — Show Your Work (Questions 9–14)

9. $\text{pK}_a = 4.74$; $\text{pH} = 4.74 + \log(0.40/0.40) = 4.74$

10. $\text{pK}_a(\text{NH}_4^+) = 9.26$; $\text{pH} = 9.26 + \log(0.25/0.35) = 9.11$

11. Initial $\text{pH} = 3.17$. After HCl : $[\text{HF}] = 0.31$, $[\text{F}^-] = 0.29$. $\text{pH} = 3.17 + \log(0.29/0.31) = 3.14$.

12. $\text{pH} = \text{pK}_a = 4.74$ (half the acid converted to acetate; $[\text{HA}] = [\text{A}^-]$)

13. $[\text{CH}_3\text{COO}^-] = 0.0040/0.080 = 0.050 \text{ M}$; $\text{K}_b = 5.6 \times 10^{-10}$; $[\text{OH}^-] = 5.3 \times 10^{-6} \text{ M}$; $\text{pOH} = 5.28$; $\text{pH} = 8.72$

14. Excess $\text{OH}^- = 0.001 \times 0.100 = 1.0 \times 10^{-4} \text{ mol}$ in $0.081 \text{ L} \rightarrow [\text{OH}^-] = 1.2 \times 10^{-3} \text{ M}$; $\text{pOH} = 2.91$; $\text{pH} = 11.09$

Short Answer & Think (Questions 15–18)

15. With $[\text{HA}] = [\text{A}^-]$, the buffer has equal amounts of both partners, so it can neutralize added acid and added base equally well — capacity is maximized and $\text{pH} = \text{pK}_a$.

16. FeCl_3 : acidic (Fe^{3+} hydrolyzes). K_2CO_3 : basic (CO_3^{2-} is a weak base). NH_4Br : acidic (NH_4^+). KClO_4 : neutral (both ions neutral).

17. The salt formed contains the conjugate acid of the weak base (e.g., NH_4^+), which hydrolyzes to produce H_3O^+ , making the solution acidic.

18. An indicator is a weak acid HIn whose color depends on the ratio $[\text{In}^-]/[\text{HIn}]$; that ratio changes gradually with pH (over roughly $\text{pK}_a \pm 1$), so the visible color shifts across a range.